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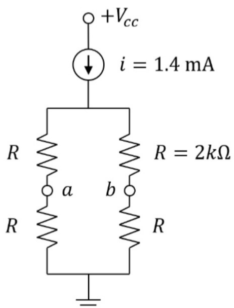
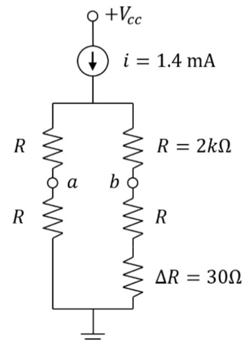
Pre-Lab Questions

For each of the four circuits shown below compute the voltages v_a, v_b, v_{cm}, and v_{dm}, where v_{cm} and v_{dm} is the common mode voltage and differential mode voltage between v_a and v_b, respectively.

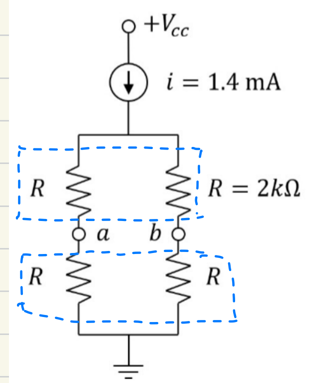
1.)



2.)



1)



$i = \frac{V_a}{R} \Rightarrow \frac{V_a}{R \parallel R}$

$V_a = i\,(R \parallel R)$

$V_a = 1.4\,mA\,(2k\Omega \parallel 2k\Omega)$

$V_a = 1.4\,\cancel{m}A\,(1\,\cancel{k}\Omega)$

$V_a = 1.4\,V$

$a = b$

$V_a = 1.4V$ $V_b = 1.4V$

$V_{cm} = \frac{V_1 + V_2}{2} = \frac{V_a + V_b}{2} = \frac{1.4V + 1.4V}{2} = \frac{2.8V}{2} = 1.4V$ $V_{cm} = 1.4V$

$V_{dm} = \frac{V_1 - V_2}{2} = \frac{V_a - V_b}{2} = \frac{1.4V - 1.4V}{2} = \frac{0}{2} = 0V$ $V_{dm} = 0V$

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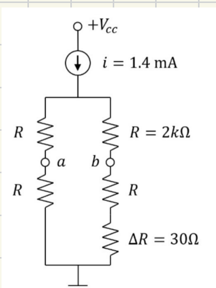
2)



$V_a = i\,(R \parallel R)$

$= 1.4\,mA\,(2k\Omega \parallel 2k\Omega)$

$= 1.4\,\cancel{m}A\,(1\,\cancel{k}\Omega)$

$V_a = 1.4V$

$V_b \neq V_a$ $V_b > V_a$

$V_b = i\,(R \parallel R + \Delta R)$

$V_b = 1.4\,mA\,(2k\Omega \parallel 2k\Omega + 30\Omega) \rightarrow \left(\frac{1}{R} + \frac{1}{R+\Delta R}\right)^{-1} \rightarrow \frac{R \cdot R+\Delta R}{R+(R+\Delta R)}$

$V_b = 1.4\,mA\left(\frac{(2k\Omega)(2030\Omega)}{2k\Omega + 2030\Omega}\right)$

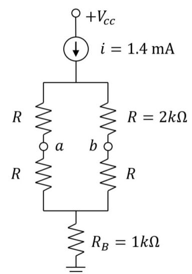
$V_b = 1.4\,mA\left[\frac{4.06 \cdot 10^6}{4.03 \cdot 10^3}\right]$

$V_b = 1.4104\,V \approx 1.41\,V$ $V_b = 1.41V$

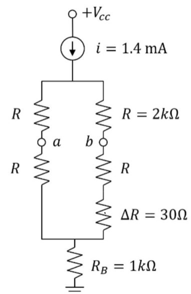
$$V_{cm} = \frac{V_1 + V_2}{2} = \frac{V_a + V_b}{2} = \frac{1.4V + 1.41V}{2} = \frac{2.81}{2} = 1.405\,V \qquad \underline{V_{cm} = 1.405\,V}$$

$$V_{dm} = \frac{V_1 - V_2}{2} = \frac{V_a - V_b}{2} = \frac{1.4V - 1.41V}{2} = \frac{-0.01}{2} = -0.005\,V \qquad \underline{V_{dm} = -5\,mV}$$

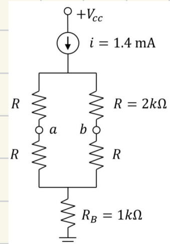
3.)



4.)



3)



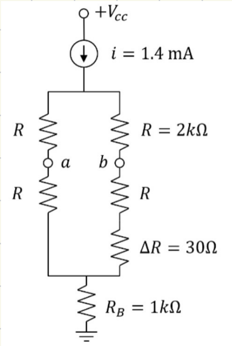
$$V_a = I R$$
$$= 1.4mA\,[2k\Omega \,||\, 2k\Omega]$$
$$= 1.4\cancel{m}A\,[1\,\cancel{k}\Omega]$$
$$\underline{V_a = 1.4\,V}$$

$$V_b = I R$$
$$= 1.4mA\,[2k\Omega \,||\, 2k\Omega + 1k\Omega]$$
$$= 1.4mA\,[2k\Omega \,||\, 3k\Omega]$$
$$= 1.4\,mA\left[\frac{3{,}000\Omega \cdot 2{,}000\Omega}{5{,}000\Omega}\right]$$
$$= 1.4\cancel{m}A\,[1.2\,\cancel{k}\Omega]$$
$$\underline{V_b = 1.68\,V}$$

$$V_{cm} = \frac{V_1 + V_2}{2} = \frac{V_a + V_b}{2} = \frac{1.4V + 1.68V}{2} = \frac{3.08V}{2} = 1.54V \qquad \underline{V_{cm} = 1.54\,V}$$

$$V_{dm} = \frac{V_1 - V_2}{2} = \frac{V_a - V_b}{2} = \frac{1.4V - 1.68V}{2} = \frac{-0.28V}{2} = -0.14V \qquad \underline{V_{dm} = -0.14\,V}$$

4)



$$V_a = 1.4mA\,(2k\Omega \parallel 2k\Omega)$$
$$= 1.4mA\,(1k\Omega)$$
$$= \underline{1.4\,V}$$
$$\underline{V_a = 1.4V}$$

$$V_b = 1.4mA\,(2k\Omega \parallel 2030\Omega)$$
$$= 1.4mA\left(\frac{2{,}000 \cdot 2030}{2000 + 2{,}030}\right)$$
$$= 1.41V$$

$$\frac{1.41V}{1K\Omega} = i \text{ [from } R_B]\qquad i = 1.41mA$$

$$V_D = 1.41\,mA\,[1K\Omega) + 1.41V = \underline{2.82V} \rightarrow \underline{V_D = 2.82V}$$

$$V_{cm} = \frac{1.4V + 2.82V}{2} = \frac{4.22V}{2} = \underline{2.11\,V}$$

$$V_{dm} = \frac{1.4V - 2.82V}{2} = \frac{-1.42}{2} = \underline{-0.71\,V}$$